

SQL Master Class: AI-Powered SQL Learner Platform

CS5200 Final Project Demo

Group 7

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Introduction

- Role-based SQL learning platform for Students, Instructors, and Admins
- Real dataset + instant autograding + AI tutor
- Stores submissions and generates individual + class analytics

Executive Summary & Milestones

- Phase I: Requirements, use cases, and initial architecture
- Phase II: EER model, relational & logical design, and analytics queries with relational algebra
- Phase III: Cloud database setup on GCP
- Phase IV: Database schema implementation, data population, and representative analytics queries

System Architecture

- Web-based front-end with role-based navigation
- Backend APIs for auth, problem bank, grading, analytics
- MySQL database for persistent learning data
- LLM-powered Open SQL for chat tutoring and NL-to-SQL analytics queries

- Dynamic SQL: analytics filters, leaderboard, submission history, role-scoped queries
- Open SQL: student & instructor chat assistants convert natural language to SQL
- Chat supports safe read queries and controlled updates where applicable

Demo !

Login Page

SQL Master Class by GPT-4

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SQL Master Class

Practice SQL, get instant feedback, and explore your data with an AI-powered assistant.

Welcome to SQL Master Class

Master SQL through hands-on practice with curated problems, real datasets, and AI-powered assistance.

[Sign In](#)

[Create Account](#)

For Students

- Practice with LeetCode-style SQL problems
- Get instant feedback and scoring
- Track your progress with analytics
- Chat with AI assistant for help

For Instructors

- Monitor student performance
- View class-level analytics
- Query student data with AI
- Track engagement metrics

For Admins

- Manage users and roles
- Content management
- System administration

Features



Real Datasets

Work with actual database schemas



Instant Feedback

Execute queries and see results immediately



AI Assistant

Natural language to SQL translation



Analytics

Track your progress over time

Learner Dashboard

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[Practice](#)

[Analytics](#)

[Chat](#)

[Test Student](#)

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Learner Dashboard

Hi, Test Student — track your SQL practice and progress.

Practice SQL Problems

Solve curated LeetCode-style questions on real datasets.

[Go to Problem Bank →](#)

My Analytics

Review your last submissions, scores, and time on task.

[View Analytics](#)

Chat Assistant

Ask questions in English – let the bot turn them into SQL.

[Open Chat](#)

Learner Dashboard

Problem Bank

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Problem Bank

Choose problems by difficulty or search by title / tags. Under the hood, this is all powered by your DB.

All Difficulties



join

[Join Three Tables](#)

MEDIUM

Dataset: SQL Interview Preparation

Problem Bank

Problem Example

Find Duplicate Emails

Difficulty: easy • Dataset: LeetCode SQL Problems

Description

Write a SQL query to find all duplicate email addresses in the Person table.
Return the email addresses that appear more than once.

Table: Person

Column Name	Type
id	int
email	varchar

id is the primary key for this table.

Each row contains an email address of a user.

SQL Editor

Write your SQL solution here...

Run & Grade

When you run your query, the backend will execute it against the problem dataset and compare against the canonical solution.

Analytics - Submissions

SQL Editor

```
select * from Person
```

Run & Grade

When you run your query, the backend will execute it against the problem dataset and compare against the canonical solution.

Result

Status: **wrong_answer** • Score: 0

Query executed successfully.

Expected 1 column(s), got 2.

id	email
1	a@example.com
2	b@example.com
3	c@example.com
4	a@example.com
5	d@example.com
6	b@example.com

Analytics - Time on Task & Leaderboard

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Analytics

Review your recent submissions, leaderboard rank, and how much time you've been putting into SQL practice.

Last 10 Submissions

Problem	Status	Score	Runtime (ms)	Submitted At	SQL
Join Three Tables	wrong_answer	0	15	2025-12-07 00:15:51	View SQL
Join Three Tables	wrong_answer	0	13	2025-12-07 00:15:50	View SQL
Join Three Tables	wrong_answer	0	17	2025-12-07 00:15:49	View SQL
Join Three Tables	wrong_answer	0	13	2025-12-07 00:15:48	View SQL
Join Three Tables	wrong_answer	0	15	2025-12-07 00:15:47	View SQL
Join Three Tables	wrong_answer	0	15	2025-12-07 00:15:41	View SQL
Find Duplicate Emails	wrong_answer	0	15	2025-12-07 00:04:11	View SQL
Second Highest Salary	wrong_answer	0	13	2025-12-06 23:50:11	View SQL
Second Highest Salary	accepted	100	14	2025-12-06 23:44:18	View SQL
Second Highest Salary	accepted	100	15	2025-12-06 23:43:43	View SQL

Time on Task per Day

Date	Minutes
2025-12-07	46
2025-12-06	1386
2025-12-05	5

This view is powered by dynamic SQL aggregations over your event logs.

Leaderboard

Rank	User	Total Score	Problems Solved
1	Test Student	290	2

Student Chat Assistant

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[Test Student](#)

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Student Chat Assistant

Ask about your progress or which problems to do next. The assistant will query your submissions and leaderboard stats.

You: what's my latest submission

Assistant: Your latest submission was for the problem titled "Join Three Tables." It was submitted on December 7, 2025, and unfortunately, it received a score of 0.0 because the answer was marked as incorrect. You submitted the SQL query ``SELECT * FROM Customer``, which might not have been what was required for the problem. Would you like to review this problem and see how to approach it correctly?

Ask about your progress or which problems to solve...

Send

Natural language → SQL → results, powered by your backend.

Student Chat Assistant

Instructor Dashboard

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[Chat Assistant](#)

Test Instructor

Logout

Instructor Dashboard

Hi, Test Instructor — monitor learner performance and explore cohort insights.

Class Analytics

View student progress, problem performance, time on task, and leaderboard rankings. Powered by dynamic SQL aggregations over your teaching data.

[Open Analytics Dashboard →](#)

Instructor Chat Assistant

Ask questions like "Which students are struggling with window functions?" or "Show me students who haven't submitted anything this week" in plain English. The AI assistant will translate your questions into SQL and provide insights.

[Open Chat Assistant →](#)

Quick Stats

9

TOTAL STUDENTS

10

ACTIVE PROBLEMS

35.3%

AVG. COMPLETION

Visit Analytics for detailed breakdowns and visualizations.

Class Analytics

Class Analytics

Monitor your class performance, pass rates, and which students are putting in the most time.

Quick Stats

TOTAL STUDENTS

9

PROBLEMS

10

AVG COMPLETION RATE

35.3%

Problem Pass Rates

Problem	Attempts	Accepted	Pass Rate	Avg Score
Find Duplicate Emails	9	2	22.2%	21.1
Second Highest Salary	8	3	37.5%	44.4
Join Three Tables	8	1	12.5%	10.7
Department Top Three Salaries	2	1	50%	50
Consecutive Numbers	2	2	100%	91.9
Customer Revenue Analysis	2	2	100%	96
Rank Scores	1	0	0%	0
Employee Hierarchy Query	1	0	0%	0
Monthly Sales Trends	1	1	100%	100

Time-on-Task Leaderboard

Student	Submissions	Estimated Minutes
Jack Taylor	2	3309
Grace Lee	2	3304
Eve Wilson	3	3304
Alice Smith	4	3304
Test Student	17	1674

Instructor Chat Assistant

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Instructor Chat Assistant

Ask cohort-level questions in English. The assistant will translate them into SQL over your teaching data.

You: who works the hardest?

Assistant: Based on the analytics, the top five students who have dedicated the most time are: 1. **Jack Taylor**: 3309 minutes, 2 submissions 2. **Grace Lee**: 3304 minutes, 2 submissions 3. **Eve Wilson**: 3304 minutes, 3 submissions 4. **Alice Smith**: 3304 minutes, 4 submissions 5. **Test Student**: 1674 minutes, 17 submissions From this data, **Jack Taylor** appears to be working the hardest in terms of total estimated time spent on the platform. However, it's also notable that **Test Student** has a high submission count, which might indicate consistent engagement, though their total time is significantly lower. If you're measuring "hard work" solely by time, then Jack Taylor wins; if you consider submissions as a metric as well, it's worth noting the contributions of the other students.

Ask about class analytics or struggling students...

Send

Natural language → SQL → results, powered by your backend.

Admin Dashboard

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[Admin Panel](#)

[Problems](#)

[Admin User](#)

[Logout](#)

Admin Dashboard

Hi, Admin User — manage users and monitor platform activity.

User Management

15 users

ID	Name	Email	Role	Created	Actions	
15	Qiannan Shi	shi.qiann@northeastern.edu	STUDENT	12/4/2025	Edit Role	Delete
14	xijia	xijia_zeng@icloud.com	STUDENT	12/4/2025	Edit Role	Delete
11	Admin User	admin@example.com	ADMIN	12/4/2025	Edit Role	Delete
12	Test Student	student@test.com	STUDENT	12/4/2025	Edit Role	Delete
13	Test Instructor	instructor@test.com	INSTRUCTOR	12/4/2025	Edit Role	Delete
1	Alice Smith	alice.smith@mit.edu	ADMIN	12/3/2025	Edit Role	Delete
2	Bob Jones	bob.jones@stanford.edu	ADMIN	12/3/2025	Edit Role	Delete
3	Charlie Brown	charlie.brown@harvard.edu	STUDENT	12/3/2025	Edit Role	Delete
4	Diana Prince	diana.prince@berkeley.edu	STUDENT	12/3/2025	Edit Role	Delete
5	Eve Wilson	eve.wilson@cmu.edu	STUDENT	12/3/2025	Edit Role	Delete
6	Frank Garcia	frank.garcia@cornell.edu	ADMIN	12/3/2025	Edit Role	Delete
7	Grace Lee	grace.lee@yale.edu	STUDENT	12/3/2025	Edit Role	Delete
8	Henry Davis	henry.davis@princeton.edu	STUDENT	12/3/2025	Edit Role	Delete
9	Iris Martin	iris.martin@columbia.edu	ADMIN	12/3/2025	Edit Role	Delete
10	Jack Taylor	jack.taylor@upenn.edu	STUDENT	12/3/2025	Edit Role	Delete

Problem Bank

Browse and manage all active problems in the system.

[View Problems](#)

Conclusion

- Delivered a complete RBAC SQL learning platform app with autograding and analytics
- Integrated LLM tutoring and natural-language query support
- Established a foundation for scalable content and richer instructional insights